Multimedia Signal Processing - 2nd Module

January, 23 2018

1. **Question (8 pts)**
   Consider two sources X and Y that emit symbols according to the following joint p.m.f.

	y_1	y_2	y_3	y_4
x_1	1/8	0	1/8	0
x_2	0	0	1/4	0
x_3	0	1/4	1/8	0
x_4	0	0	0	1/8

   (a) Compute the probability mass functions of symbols of X and Y, separately.

   (b) Compute the entropy of both sources, i.e., $H(X)$ and $H(Y)$.

   (c) Compute the joint entropy $H(X, Y)$.

   (d) Design the Huffman code to encode single symbols of X and compare the average code length to $H(X)$.

   (e) Design the Huffman code to jointly encode symbols of X and Y and compare the average code length to $H(X, Y)$.

2. **Question (5 pts)**
   (a) Illustrate the block diagram of a typical transform coding scheme. Quickly comment on the role of each block (i.e., what do they do and what is their purpose), and specify at which point of the scheme we obtain the encoded and the decoded signals.

   (b) Report pros and cons of using DCT or KLT transform.

3. **Question (5 pts)**
   (a) Describe the source-system model used to synthesize speech signals.

   (b) Explain how a linear predictive vocoder exploits this scheme, and which are the main parameters sent from the encoder to the decoder.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**

   (a) Consider a typical hybrid DPCM/DCT video coding scheme.

   - [T] [F] Block-wise transform is bound to work on 8×8 pixel blocks by every encoder.
   - [T] [F] The decoder computes motion vectors using the same "motion estimation" block used by the encoder.
   - [T] [F] Other frames in addition to the first one are intra-coded.
   - [T] [F] The displaced frame difference should in general contain less energy than a simple frame difference if motion estimation and compensation are properly performed.

   (b) Consider two discrete memoryless sources X and Y emitting symbols from a finite alphabet.

   - [T] [F] It is possible to encode symbols from X with distortion higher than the source variance σ_X^2 according to rate-distortion theory.
   - [T] [F] Mutual information $I(X;Y)$ must be strictly positive for correlated sources.
   - [T] [F] The maximum entropy $H(X)$ does not depend on the size of the alphabet.

   (c) Consider a closed-loop DPCM.

   - [T] [F] Given a fixed rate, DPCM always provides less distortion than simple PCM.
   - [T] [F] The use of a closed-loop architecture avoid the drift phenomenon typical of open-loop one.
   - [T] [F] The reconstruction error equals the quantization error introduced on the residual.

5. **Question (Matlab - 10pts)**

   You are called as a signal processing expert to assess which image, in a set of two, has undergone lossy compression. Unfortunately, both files at hand are saved as lossless PNG images. Therefore, file format is not helpful in solving the issue. As you know that the most widespread lossy compression format for digital images is JPEG, you setup a test in order to determine which image was, at some point in time, JPEG compressed. Indeed, you know that the distribution of some DCT coefficients is affected by the JPEG quantization step, so you expect the histogram of some DCT coefficients to be very different between compressed and uncompressed images.

   (a) Load `img1.png`, display it on screen, and compute the block-wise 8×8 DCT for each block (hint: the image is one-channel only).

   (b) Compute the histogram of the DCT coefficients at spacial frequency $(3, 3)$. Consider only bins centered in $[-20 : 20]$, and round DCT coefficients to the nearest integer before computing the histogram.

   (c) Repeat steps (a) and (b) on `img2.png`, then plot the two histograms.

   (d) Comment on the obtained histograms: what do you observe? do you have any clue about which image has been JPEG compressed in the past? why?

   To automatize the process, you decide to compare the amount of energy at high frequencies of the two histograms.

   (e) Compute the absolute value of the 1D DCT (hint: `dct`) of the two histograms and plot them. How would you extract from this a single number for each image that you could use as an indicator of JPEG compression? How would you set a threshold to asses which image is JPEG compressed?